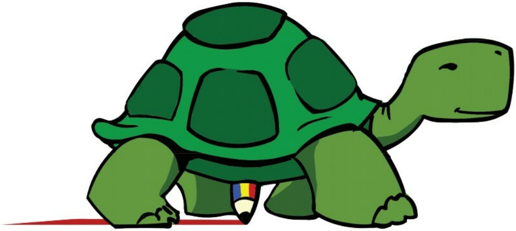
KTurtle:
A Programming Language for Children



The ideal age at which to start programming is a debatable topic. While prodigies like Zuckerberg and Musk started at a pretty early age, others who started later on in life have also done quite well. The best thing for a beginner would be an app that helps in the programming process and encourages logical and systematic thinking.

Programming is often considered to be a difficult task, to be mastered only by adults and definitely not children. Of course, there are a lot of exceptions—programming child prodigies like Mark Zuckerberg and Elon Musk mastered the art of programming at a very tender age. Zuckerberg started programming at a young age and ended up creating Facebook. Similarly, Elon Musk, the renowned entrepreneur, inventor and futurist, was selling video games he'd developed himself in his early teens. These brilliant people can't be taken as the standard while teaching programming to randomly chosen children.

I started learning programming when I was 15, and was quite okay with the idea of using computers to do something other than playing video games. Personal experiences aside, the important question is, 'What is the right age to start learning programming?' The answer is sure to be contested. I couldn't find any scientific studies regarding this. All I could find were a bunch of articles sharing personal opinions.

A child aged eight to ten years can start learning programming. By that age, a child knows all the basic arithmetic necessary to do some sort of programming. The other important question is 'What should be taught as the first programming language?' Should it be a normal programming language like Python or Java, or should it be a programming language specifically designed to teach programming principles to children. Well, if you are a 15-year-old with good mathematical skills then start learning Python immediately. In my opinion it will be slightly easier for a first-time learner to start with a language like Python that has its own shell, rather than languages like C or Java. But an eight-year-old child who wants to learn programming should explore Logo, a programming language specifically designed for kids.

Logo can be called a general-purpose programming language. It is described as a dialect of Lisp, a functional programming language. In a very a strict sense, Logo is not a programming language. It is just a programming language specification. Logo has conditional statements, loops, functions, recursion, file handling, etc, which are necessary parts of any general-purpose programming language. These constructs can be used to train children in programming. The most important aspect of Logo, and what makes it a very useful training tool, is its turtle graphics.

Turtle graphics are vector graphics using a relative cursor on a Cartesian plane. Although this definition is very precise, it may not give you a good idea about turtle graphics. In simple terms, turtle graphics is used to draw graphic images using programs. So, essentially, you can write a program in Logo, which when executed will draw a graphical image on your screen. A 'turtle' in turtle graphics is a cursor, which